

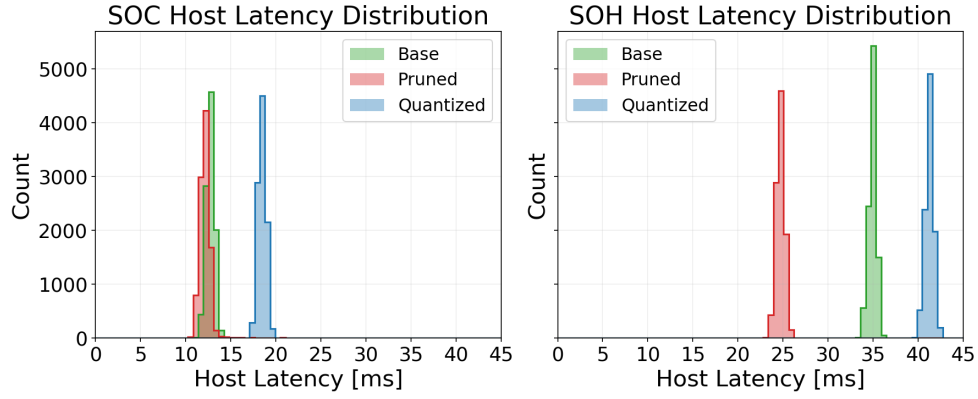


Figure 7.13: End-to-end host-latency distributions for repeated STM32H753ZI streaming measurements. Each observation includes UART communication and the on-device estimator update. The larger absolute Base–Pruned separation for SOH reflects the larger recurrent kernel, while the Quantized variants include the overhead of the mixed-precision weight reconstruction.

Table 7.3: Composite pruning and quantization summary. Values are reported as SOC/SOH. ΔMAE is given in percentage points of full scale. Flash, RAM, and E_{est} are relative changes with respect to the corresponding Base model. The utility score uses the equal weights in Equation 7.22, with lower values indicating the more favourable combined operating point.

Variant	ΔMAE [pp]	ΔFlash [%]	ΔRAM [%]	ΔE_{est} [%]	U
Pruned	−0.35 / +0.61	−41 / −46	−18 / −20	−43 / −44	0.71 / 0.91
Quantized	+0.10 / +0.56	−50 / −59	−20 / −23	+399 / +29	1.83 / 1.03

7.12 Discussion of Embedded Compression Results

7.12.1 Compression Behaviour of SOC and SOH Estimators

Pruning and quantization produce different outcomes for the two estimation tasks. In SOC estimation, Pruned achieves an MAE that is 0.35 percentage points below Base, while Quantized is 0.11 percentage points above it. The saliency and weight-distribution analysis confirms that the specified L2 procedure removed the intended low-ranked channels. It does not establish why the Pruned model has the lower test error. Only one trained model and one predefined split are compared for each task,